Lecture 14

25 February 2026 16:24

Recap

Properties

$T = 1000$

Fixed step-size rule (with steepest decent direction):
Initialize: $x_0 \in \mathbb{R}^n$
Iterate: for $t \geq 0$, $\cancel{x_{t+1} = x_t - \alpha \nabla f(x_t)}$ where $\alpha > 0$

$x_0, x_1, \ldots x_{1000}$

For $T \geq 1$,

$$\sum_{t=0}^{T-1} \nabla f(x_t)^T (x_t - x^*) \leq \frac{\alpha}{2} \sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2 + \frac{1}{2\alpha} \|x_0 - x^*\|^2$$

Theorem: (a) If f is Lipschitz continuous with parameter B then using

$$\alpha = \frac{\|x_0 - x^*\|}{B\sqrt{T}}$$

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$$\cancel{\frac{1}{T}} \sum_{t=0}^{T-1} \nabla f(x_t)^T (x_t - x^*) \leq \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$

(b) If f is also convex then

$$\frac{1}{T} \sum_{t=0}^{T-1} (f(x_t) - f(x^*)) \leq \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$

* we must fix the step size.
- what if we do not fix T in beginning.
  - we can use suitably designed α_t, $t \geq 0$

* $f(y) - f(x) \leq B\|y - x\|$ $\forall x, y$.
- if B is a Lipschitz constant then any $\bar{B} \geq B$ is also a Lipschitz constant.
  - we can use any Lipschitz const, but a larger one requires more iterations

(b) implies $\min_{0 \leq t \leq T-1} f(x_t) - f(x^*) \leq \frac{\|x_0 - x^*\| B}{\sqrt{T}}$

suppose $x^{(t)} = \underset{0 \leq l \leq t}{\operatorname{argmin}} f(x_l)$

(best point upto t)

$$f(x^{T-1}) - f(x^*) \leq \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$

suppose we want L.H.S to be $\leq \epsilon$. $(f(x^{T-1}) - f(x^*) \leq \epsilon)$

we must choose T s.t. $\frac{\|x_0 - x^*\| B}{\sqrt{T}} \leq \epsilon$

$$\Rightarrow T \geq \left(\frac{\|x_0 - x^*\| B}{\epsilon}\right)^2 \quad \left(T \geq \frac{C_1}{\epsilon^2}\right)$$

- larger $B \Rightarrow$ higher T

Theorem: (a) If $\cancel{f \text{ is } L\text{-smooth}}$ then for $\alpha < \frac{2}{L}$, (∇f is Lipschitz cts with parameter L)

$$\lim_{t \to \infty} \nabla f(x_t) = 0$$

(b) If f is also convex then using $\alpha = \frac{1}{L}$,

$x_1 - x_2 - x_3 - \ldots$
$\nabla f(x_1)$ $\nabla f(x_2)$ $\nabla f(x_3)$ $\rightarrow$

$f(x_T) - f(x^*) \le \frac{L}{2T}\|x_0 - x^*\|^2$ $\quad f(x_T) - f(x^*) \le \epsilon$

suppose we choose T such that

$$\frac{L}{2T}\|x_0 - x^*\|^2 \le \epsilon$$

$$T \ge \frac{L\|x_0 - x^*\|^2}{2\epsilon} \qquad \left(T \ge \frac{C_2}{\epsilon}\right)$$

Theorem: If f is L-smooth and strongly convex with parameter m then using $\alpha = \frac{1}{L}$,

$\left(1 - \frac{m}{L}\right)^T \downarrow T$

$(0.9)^T$

1. $\|x_T - x^*\|^2 \le \left(1 - \frac{m}{L}\right)^T \|x_0 - x^*\|^2$,
2. $f(x_T) - f(x^*) \le \frac{L}{2}\left(1 - \frac{m}{L}\right)^T \|x_0 - x^*\|^2$.

$=: \beta$

suppose we want $f(x_T) - f(x^*) \le \epsilon$

$$\frac{L}{2}\beta^T\|x_0 - x^*\|^2 \le \epsilon \qquad \left(C_3 = \frac{L}{2}\|x_0 - x^*\|^2\right)$$

$$\beta^T \le \frac{\epsilon}{C_3}$$

$$T \ln \beta \le \ln\left(\frac{\epsilon}{C_3}\right)$$

$$T \ge \frac{-\ln(\epsilon/C_3)}{-\ln\beta} = \frac{\ln(C_3/\epsilon)}{\ln(1/\beta)} = \frac{\ln(C_3/\epsilon)}{\ln\left(\frac{1}{1-m/L}\right)}$$

$\frac{m}{L} \le 1$

$$T \ge O(\ln(1/\epsilon))$$

$\epsilon = 10^{-2}$

$\frac{1}{\epsilon} = 10^2$, $\frac{1}{\epsilon^2} = 10^4$

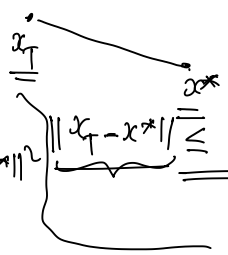
$\ln(1/\epsilon) = \ln(100)$

$\epsilon < 1/\epsilon$

Theorem: If f is L-smooth and strongly convex with parameter m then using $\alpha = \frac{1}{m+L}$,

1. $\|x_T - x^*\|^2 \le \left(\frac{L-m}{L+m}\right)^{2T}\|x_0 - x^*\|^2$,
2. $f(x_T) - f(x^*) \le \cancel{\frac{L}{2}\left(\frac{L-m}{L+m}\right)^{2T}}\|x_0 - x^*\|^2$ $\quad f(x_T) - f(x^*) \le \epsilon$

suppose



$\|x_T - x^*\| \le$?

$$f(x_T) \ge f(x^*) + \underbrace{\cancel{\nabla f(x^*)^T(x_T - x^*)}}_{=0} + \frac{m}{2}\|x_T - x^*\|^2$$

$$\frac{m}{2}\|x_T - x^*\|^2 \le f(x_T) - f(x^*) \le \epsilon$$

$$\Rightarrow \|x_T - x^*\| \le \sqrt{\frac{2\epsilon}{m}}$$

$\frac{L}{m} > 1$

- $\frac{L}{m}$ is referred to as the condition number of f.
- In many practical problems we do not know L and m. In these cases we may use an upper bound on L and a lower bound on m.

$L' \ge L$, $m' \le m$

we can use L', m'

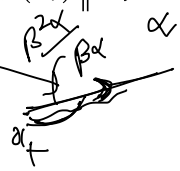
Comparison (objective: $\min_{1 \le t \le T}(f(x_T) - f(x^*)) \le \epsilon$)

f (convex)	Number of iterations (T)
~~Lipschitz~~	$\Omega\left(\frac{1}{\sqrt{\epsilon}}\right)$
Smooth ~~(Lipschitz~~ gradient)	$\Omega\left(\frac{1}{\epsilon}\right)$
Smooth an~~d strongly c~~onvex	$\Omega\left(log\left(\frac{1}{\epsilon}\right)\right)$

Armijo's rule (with steepest descent direction):
Initialize: $x_0 \in \mathbb{R}^n$
Iterate: for $t \geq 0$, $\quad x_{t+1} = x_t - \alpha_t \nabla f(x_t)$ where $\alpha_t > 0$

- Backtracking line search (Armijo's rule): Fix $\alpha > 0$ and $\beta, \gamma \in (0,1)$. Given x_t, d_t,
  Initialize: $m = 0$
  While $f(x_t - \beta^m \alpha \nabla f(x_t)) > f(x_t) - \sigma \beta^m \alpha \|\nabla f(x_t)\|^2$
  $m = m + 1$
  End
  $\alpha_t = \beta^m \alpha$

$d_t = -\nabla f(x_t)$



on any iteration, we stop backtracking when

$$f(x_t - \beta^m \alpha \nabla f(x_t)) \leq f(x_t) - \sigma \beta^m \alpha \|\nabla f(x_t)\|^2$$

what ensures that we will ever stop.

Descent lemma

$$\underbrace{f(x_t - \beta^m \alpha \nabla f(x_t))}_{=:y} \leq f(x_t) - \nabla f(x_t)^T (\alpha \beta^m \nabla f(x_t)) + \frac{L}{2} \|\nabla f(x_t)\|^2 \beta^{2m} \alpha^2$$

$$\left(f(y) \leq f(x_t) + \nabla f(x_t)^T (y - x_t) + \frac{L}{2} \|y - x_t\|^2 \right)$$

$$= f(x_t) - \alpha \beta^m \left(1 - \frac{L}{2} \alpha \beta^m\right) \|\nabla f(x_t)\|^2$$

Armijo's rule's iterations will surely stop if

$$\cancel{f(x_t)} - \cancel{\alpha \beta^m} \left(1 - \frac{L}{2} \alpha \beta^m\right) \cancel{\|\nabla f(x_t)\|^2} \leq \cancel{f(x_t)} - \cancel{\sigma \beta^m \alpha \|\nabla f(x_t)\|^2}$$

$$\Leftrightarrow \quad 1 - \frac{L}{2} \alpha \beta^m \geq \sigma$$

$$\Leftrightarrow \quad \frac{L}{2} \alpha \beta^m \leq 1 - \sigma$$

$$\Leftrightarrow \quad \alpha \beta^m \leq \frac{2(1-\sigma)}{L}$$

So, we are guaranteed that

$$\alpha_t \geq \frac{2\beta(1-\sigma)}{L}$$

Lemma: If f is L-smooth then step-size adjustment is guaranteed to terminate before $\beta^m \alpha$ goes below $\frac{2\beta(1-\sigma)}{L}$. So, $\alpha_t \geq min\left\{\alpha, \frac{2\beta(1-\sigma)}{L}\right\}$.

Theorem: If f is L-smooth then $\lim_{t \to \infty} \nabla f(x_t) = 0$.

Theorem: If f is L-smooth and strongly convex with parameter m and $\alpha = 1$ then

$$f(x_T) - f(x^*) \le \left(1 - \min\left\{2\sigma m, \frac{2\beta\sigma m}{L}\right\}\right)^T (f(x_0) - f(x^*)).$$

So far: 1st order methods (depending on gradients)

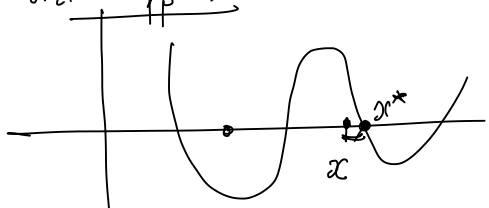
Now, 2nd order methods, (use Hessian)

* goal: to obtain x^* s.t. $\nabla f(x^*) = 0$

· Consider $g(x)$

want to solve $g(x) = 0$

if g is complicated we can use its first order approx.



$x \approx x^*$

$$g(y) \approx \underbrace{g(x) + \nabla g(x)^T (y - x)}_{\bar{g}(y)}$$

$g(y) \approx \bar{g}(y)$ in the neighborhood of x

- try to find y s.t. $\bar{g}(y) = 0$

$y \approx x^*$ if x is in the neighborhood of x^*

Suppose we are at x_t.

- want to find x^* s.t. $\nabla f(x^*) = 0$

First order approximation of $\nabla f(x)$ around x_t

$$\nabla f(x_t) + \nabla^2 f(x_t)(y - x_t) =$$

set it to 0.

$$y - x_t = -\nabla^2 f(x_t)^{-1} \nabla f(x_t)$$

$$\Rightarrow y = \underbrace{x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)}_{\text{call it } x_{t+1}}$$

and continue.

$$x_{t+1} = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

(Newton's method)

Another motivation

2nd order approximation of f around x_t

$$\min_{x \in \mathbb{R}^n} f(x)$$

$$\min \quad f(x_t) + \nabla f(x_t)^T (x - x_t) + \frac{1}{2}(x - x_t)^T \nabla^2 f(x_t)(x - x_t)$$

$x \in \mathbb{R}^n$

$$\nabla f(x_t) + \nabla^2 f(x_t)(x - x_t) = 0$$

$$\Rightarrow x = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$$

Newton's method

Initialize: $x_0 \in \mathbb{R}^n$

Iterate: for $t \geq 0$, $x_{t+1} = x_t - \nabla^2 f(x_t)^{-1} \nabla f(x_t)$ (assuming that $\nabla^2 f(x_t)$ is invertible)

$x_{t+1} = x_t - \alpha_t \nabla f(x_t)$

- This can be seen as a gradient method with $\alpha_t = \nabla^2 f(x_t)^{-1}$.
- Motivation: convergence will be faster by using second derivatives.

Benefit of Newton's method:

Assume that $\nabla^2 f(x): \mathbb{R}^n \to \mathbb{R}^{n \times n}$ is Lipschitz continuous with para D.

$$\|x_{t+1} - x^*\| = \|x_t - x^* - \nabla^2 f(x_t)^{-1}(\nabla f(x_t) - \underbrace{\nabla f(x^*)}_{=0})\|$$

$$= \|\nabla^2 f(x_t)^{-1}(\nabla f(x_t) - \nabla f(x^*) - \nabla^2 f(x_t)(x_t - x^*))\|$$

$$= \|\nabla^2 f(x_t)^{-1} \nabla^2 f(\underbrace{x_t + \alpha(x^* - x_t)}_{= \alpha x^* + (1-\alpha) x_t})(x_t - x^*) - \nabla^2 f(x_t)(x_t - x^*)\|$$

MVT

$$\left\{ f(x) - f(x) = \left(\nabla f(\alpha x + (1-\alpha) y)\right)^T (x - y) \right.$$

$$= \|\nabla^2 f(x_t)^{-1} (\nabla^2 f(x_t + \alpha(x^* - x_t)) - \nabla^2 f(x_t))(x_t - x^*)\|$$

$$\|ABz\| \leq \|A\| \|Bz\| \leq \|A\| \|B\| \|z\|$$

$$\text{L.H.S} \leq \|\nabla^2 f(x_t)^{-1}\| \underbrace{\|\nabla^2 f(x_t + \alpha(x^* - x_t)) - \nabla^2 f(x_t)\|}_{\leq D \alpha \|x^* - x_t\|} \|x_t - x^*\|$$

$\alpha \leq 1$

$$\leq D \|\nabla^2 f(x_t)^{-1}\| \|x^* - x_t\|^2$$

$$\leq \frac{D}{\sqrt{\lambda_{min}(\nabla^2 f(x_t))}} \|x^* - x_t\|^2$$

smallest eigenvalue modulus of $\nabla^2 f(x_t)$

$\nabla^2 f(x_t)^{-1} \nabla f(x_t)$ is PD.

iff $\nabla^2 f(x_t)$ is invertible

A symmetric

$\|A\| \leq \sqrt{\lambda_{max}(A)}$

$\lambda_{max}(A^{-1}) = \frac{1}{\lambda_{min}(A)}$

$$\|A\| = \max_{x: \|x\| = 1} \|Ax\|$$

$$= \max_{x: \|x\| = 1} \sqrt{x^T A^T A x}$$

$$= \sqrt{\lambda_{max}(A^T A)}$$

A is symmetric

$\lambda_{max}(A) = \sqrt{\lambda_{max}(A^T A)}$

$\lambda_{min}(A) = \sqrt{\lambda_{min}(A^T A)}$

Suppose A has eigenvalues $1, 0.2, -0.3, -1.2$

$\lambda_{min}(A) = -0.3$

$\lambda_{min}(A^T A) = 0.09 \Rightarrow \sqrt{\lambda_{min}(A^T A)} = 0.2$

$\sqrt{\lambda_{max}(A^T A)} = 1.2$

Theorem: If $\nabla^2 f(x^*)$ is invertible and $\nabla^2 f(x)$ is Lipschitz continuous with a parameter D in the neighbourhood of x^* then

$$\|x_{t+1} - x^*\| \le \frac{D}{\lambda(\nabla^2 f(x_t))}\|x_t - x^*\|^2$$

provided x_t is close enough to [Equation] ([Equation] denotes the smallest eigenvalue modulus of [Equation]).

first order gradient descent

$\|x_{t+1} - x^*\| \le C\|x_t - x^*\|$

In NM $\|x_{t+1} - x^*\| < \|x_t - x^*\|$ if $\frac{D}{\lambda(\nabla^2 f(x_t))}\|x_t - x^*\|^2 < \|x_t - x^*\|$

i.e. if $\frac{D}{\lambda(\nabla^2 f(x_t))}\|x_t - x^*\| < 1$

x_{t+1} x_{t+2} x^* x_t

- If $\frac{D}{\lambda(\nabla^2 f(x_t))}\|x_t - x^*\| < 1$ for all $t \ge 0$ and x_ts are close to x^* then they converges to x^*.
- Newton's method converges faster near x^* than the earlier gradient methods (order-2 convergence as opposed to order-1 convergence).

Modified Newton's method

given x_t

if $\nabla^2 f(x_t)$ is invertible and $f(x_t - \nabla^2 f(x_t)^{-1}\nabla f(x_t)) < f(x_t)$

$x_{t+1} = x_t - \nabla^2 f(x_t)^{-1}\nabla f(x_t)$

else pick δ s.t.

$\lambda_{min}(\nabla^2 f(x_t) + \delta I) \ge \Delta$ pre defined parameter (ex: $\Delta = 0.05$)

$d_t = -(\nabla^2 f(x_t) + \delta I)^{-1}\nabla f(x_t)$

then $\nabla f(x_t)^T d = -\nabla f(x_t)^T(\nabla^2 f(x_t) + \delta I)^{-1}\nabla f(x_t) < 0$

so d_t is a descent direction.

$\alpha_t = 1$

x_t

$x_{t+1} = x_t + \alpha_t d_t$ (choose α_t using Armijo's rule or fixed)

$(f(x_{t+1}) \le f(x_t))$